

Clear optics — Solution

X23 (2023) — English translation

A1 Measure the focal lengths f and F by taking the image A' on the screen. Use $H \approx 1.20$ m to measure. On the answer sheet, indicate with a cross the correspondence between the color of the lens and the names L_f and L_F .

Let's
use
the
thin
lens
for-
mula

$$\frac{1}{a} + \frac{1}{H - a} = \frac{1}{f}$$

For
each
lens
we
will
make
two
mea-
sure-
ments.

White

Lens:

Near

Fo-

cus:

$H =$

$124.2\text{cm}, \quad a =$

$7.3\text{cm} \Rightarrow$

$f =$

6.9cm

Far

Fo-

cus:

$H =$

$124.2\text{cm}, \quad a =$

$116.5\text{cm} \Rightarrow$

$f =$

7.2cm

Blue 0.20

Lens:

Near

Fo-

cus:

$H =$

$124.5\text{cm}, \quad a =$

$18.0\text{cm} \Rightarrow$

$F =$

15.4cm

Far

Fo-

cus:

$H =$

$124.2\text{cm}, \quad a =$

B1 Align the lens so that $a = a_0 + F/2$. In further paragraphs we will study the characteristic size of such images, so choose which linear size - height h or width w of the fuzzy image you will measure in subsequent paragraphs. Justify your choice and mark it with a cross on your answer sheet. This size will be designated p .

The 0.20

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source,
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we
will
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the
width.**

B2 Cover the part of the lens facing the screen with an aperture of diameter D . **0.45**
For all apertures, measure the image size p .

B3 Plot the dependence $p(D)$.

The 0.40

graph

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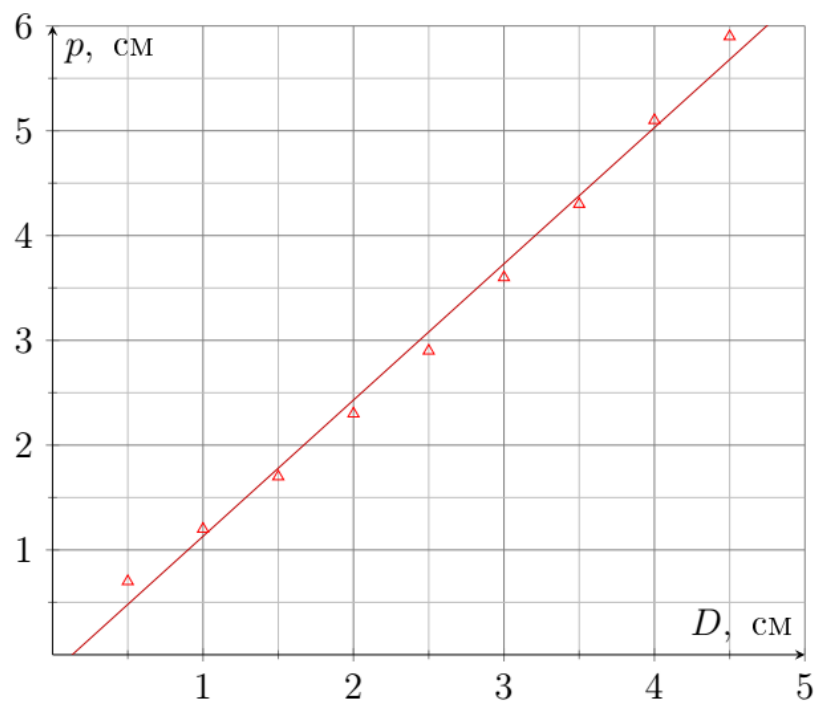
up-

ward

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vex-

ity.



B4 With the same arrangement of objects, remove the aperture from the lens and measure the size p_0 of the image on the screen. Determine the diameter D_0 of the open part of the lens based on p_0 , compare the result with direct measurements.

Linear 0.10

**de-
pen-
dence
co-
effi-
cients:**

$p =$
 $-0.17 \text{ cm} +$

$1.3 \cdot$

$D.$

**At
the
same
time**

$p_0 =$
 $7.0 \text{ cm}.$

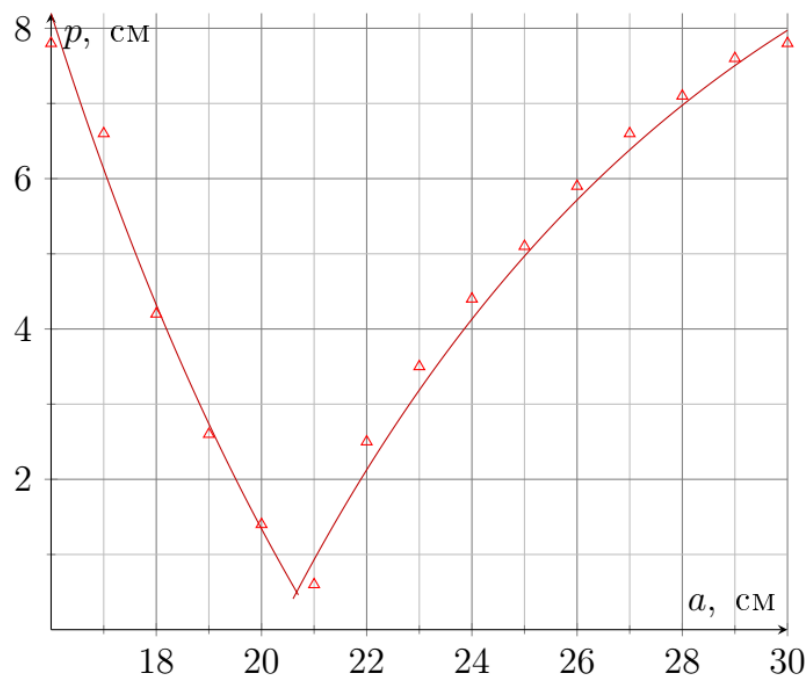
**From
this
de-
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dence**

$D_0 =$
 $5.5 \text{ cm}.$

**From
di-
rect
mea-
sure-
ments**

$D_0 =$
 5.2 cm

B5 With 11 different a in the range $[a_0 - F/3, a_0 + F/2]$, measure the size p of the image on the screen. **0.55**



B6 Construct a linearized graph of p versus a . From the graph, find the focal length F and compare it with the one obtained earlier. Consider the filament to be $0.10 \text{ cm} \times 0.30 \text{ cm}$ in size.

From
the
thin
lens
for-
mula

$$\frac{1}{b} + \frac{1}{a} = \frac{1}{F} \quad \Rightarrow \quad b = \frac{Fa}{a - F},$$

and
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size
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is

$$W_A \frac{b}{a}$$

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the
screen.

At

$$a >$$

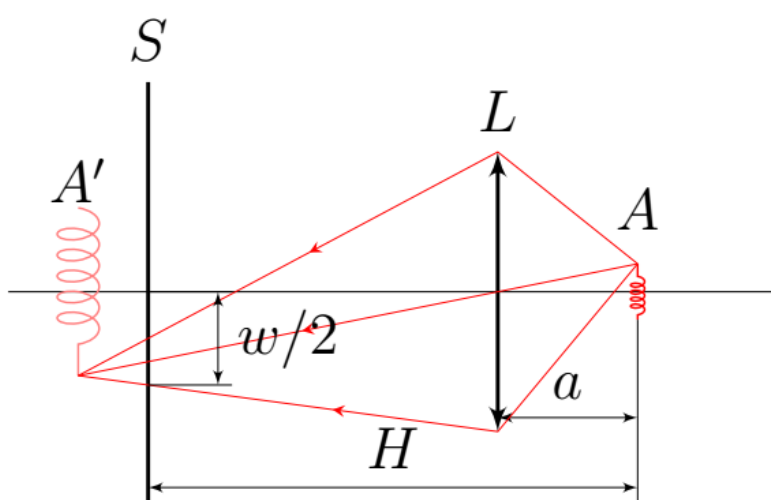
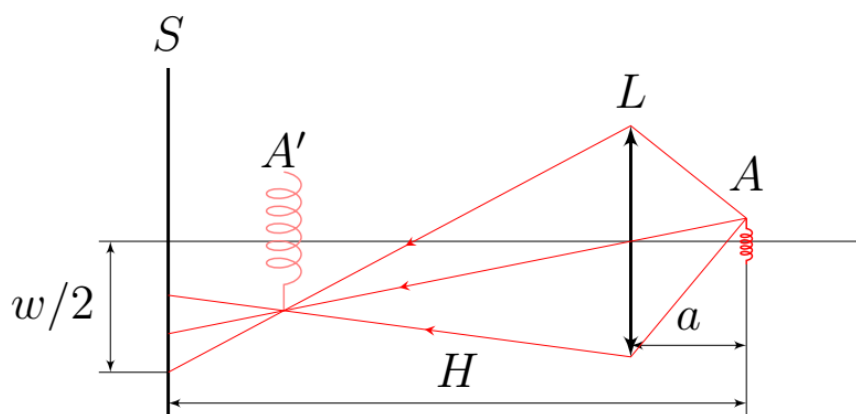
$$a_0$$

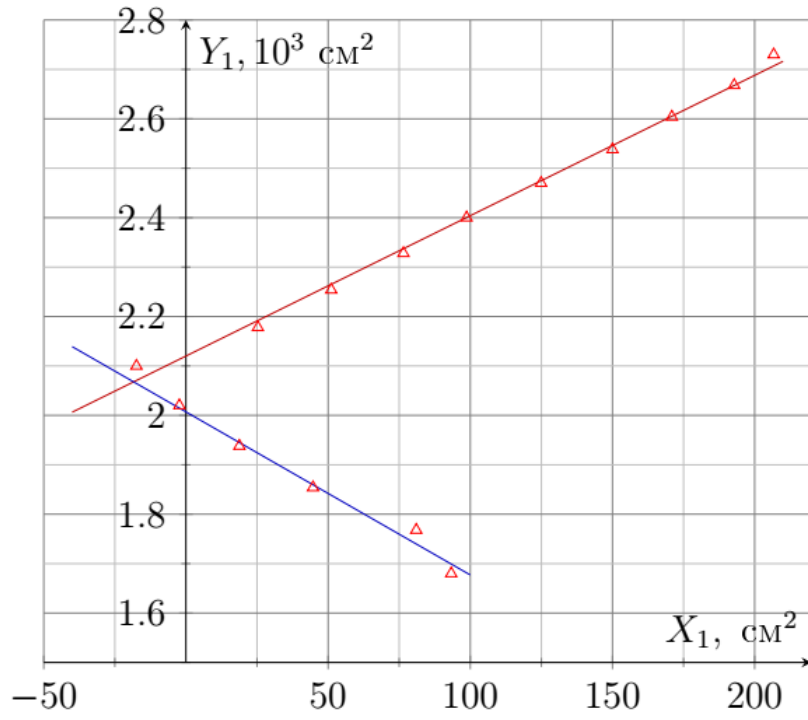
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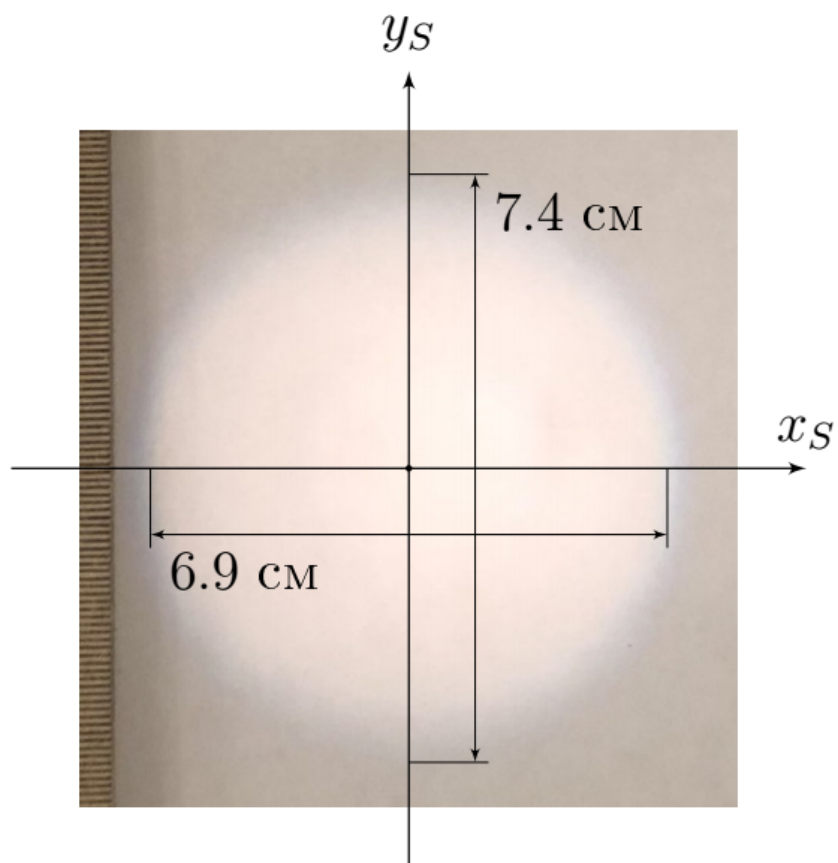
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- B7** Position the lens so that $a = a_0 + 2F/5$. Draw the border of the image on graph paper. Label the center of the image, the horizontal axis x_S and the vertical axis y_S .

0.30



B8 Calculate the position of the boundary Γ of the image relative to its center at 11 points lying in the area $x_S > 0, y_S > 0$. Place these dots on the same sheet of graph paper where you sketched the image.

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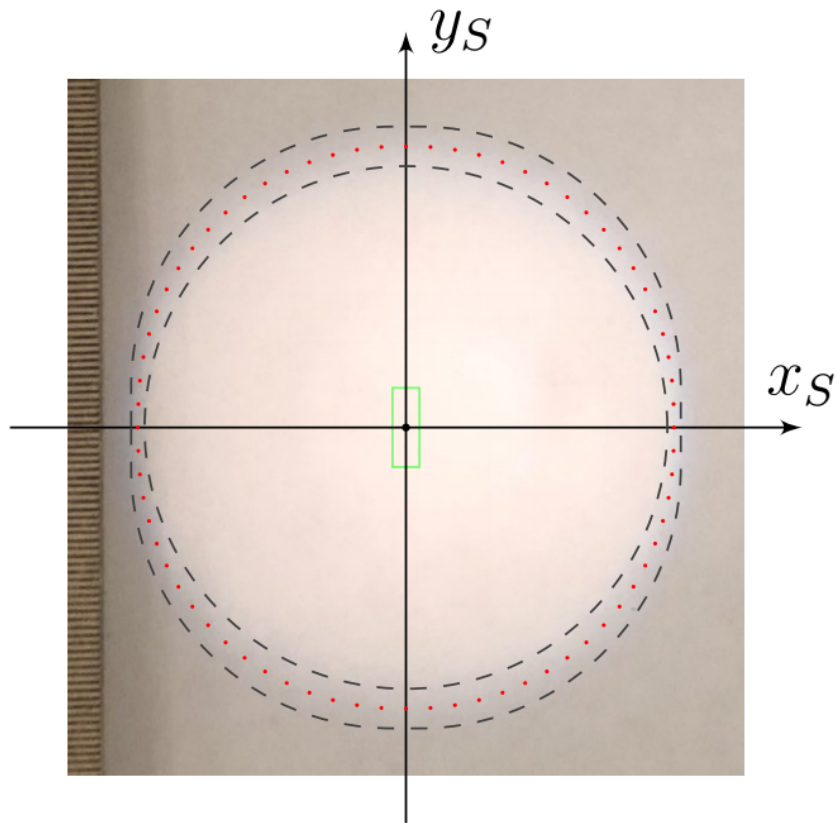
a

rect-

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C1 On your answer sheets, place a cross next to the true statements.

Answer:

L_F

Bor-

der

blue

at

$a <$

$a_0,$

red

at

$a >$

$a_0 L_F$

Bor-

der

red

at

$a <$

$a_0,$

blue

at

$a >$

$a_0 \times L_f$

Bor-

der

blue

at

$a <$

$a_0,$

red

at

$a >$

$a_0 L_f$

Red

bor-

der

for

$a <$

$a_0,$

blue

for

$a >$

$a_0 \times$

C2 On your answer sheets, mark the cross next to the true statements about n_b and n_r . Justify your choice.

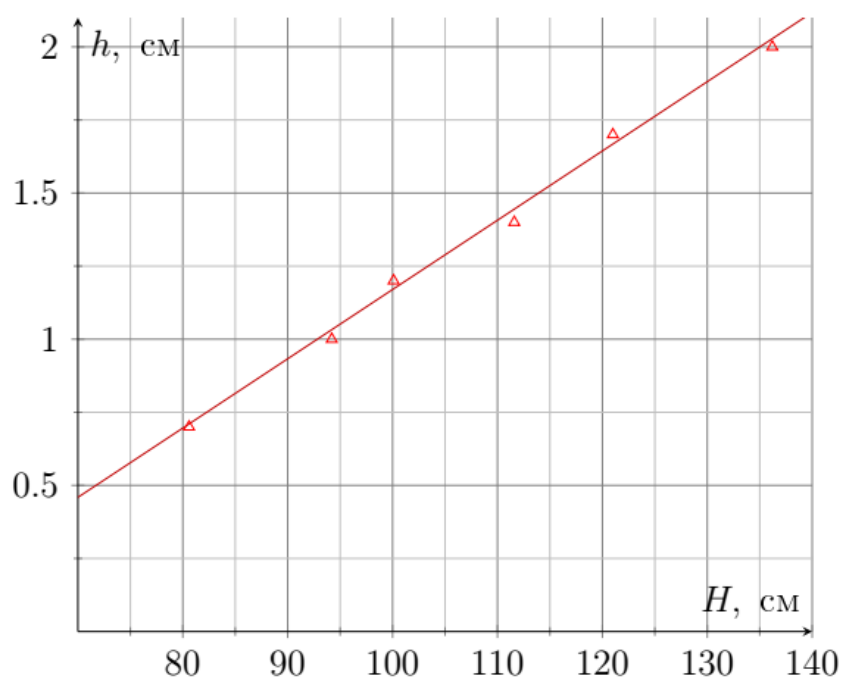
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 b_r),
which
im-
plies

$$\frac{1}{b_b} > \frac{1}{b_r} \quad \Rightarrow \quad \frac{1}{F_b} > \frac{1}{F_r},$$

which
im-
plies
the
an-
swer
from
the
grinder
for-
mula:

$$\frac{1}{F} = (n-1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

D1 With 7 different H , measure the image size p under close focus conditions. **0.70**



D2 Linearize the dependence of p on H and plot it. Calculate the source parameters W_A and H_A , making additional measurements if necessary.

The height of the focused image h is expressed through the height of the spiral H_A :

$$\frac{H_A}{a} = \frac{w}{H - a}.$$

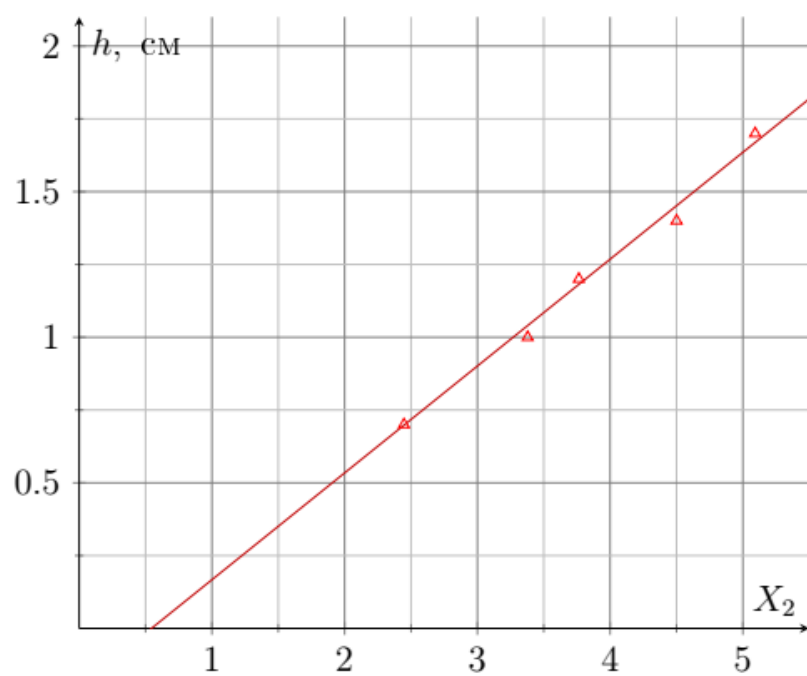
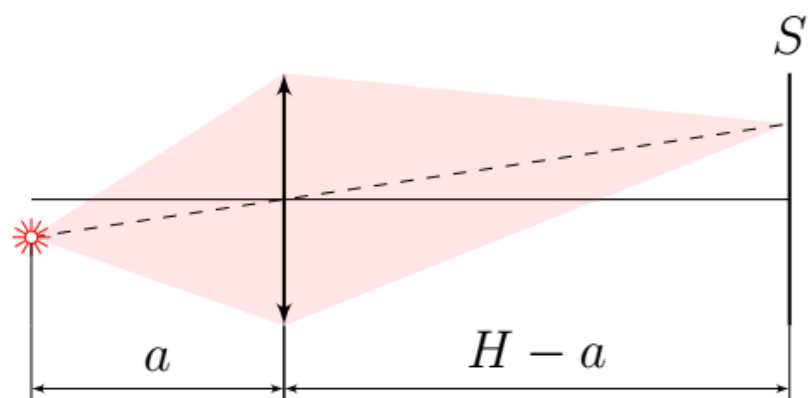
Let us find a from the thin lens formula:

$$\frac{1}{a} + \frac{1}{H - a} = \frac{1}{F} \quad \Rightarrow \quad a^2 - aH + FH = 0$$

Of the two roots, we are interested in the smaller

$$a = \frac{1}{2} \left(H - \sqrt{H^2 - 4FH} \right).$$

In this case, the de-



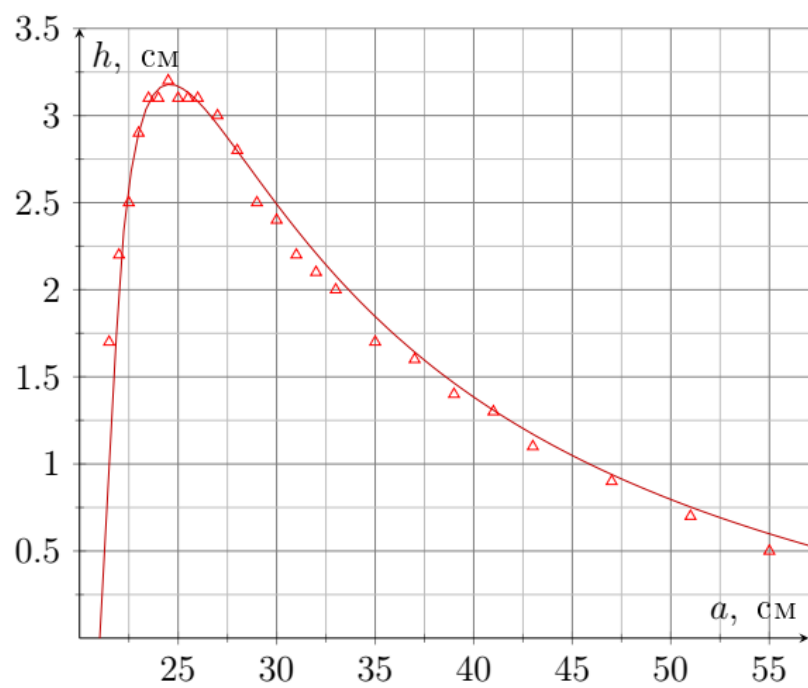
D3 Count the number of N_A spirals in the source spiral.

Answer: 10

Let's
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at
the
im-
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on
the
screen
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larged
with
a
lens,
and
we
will
also
use
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blur
due
to
aber-
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tions.

$$N_A = 18$$

E1 For various fixed values of a , select distances l that correspond to a clear image. Measure the height of the clear image h . If several different values of l produce several clear images, then select the image with the larger h . The dependence of h on a has a characteristic point, describe it and measure its surroundings especially carefully. Take at least 15 measurements in total. **0.85**



E2 Obtain a theoretical formula for the dependence of h on a . Construct a linearized graph and show that the theoretical formula is valid.

**Thin
lens
for-
mu-
las:**

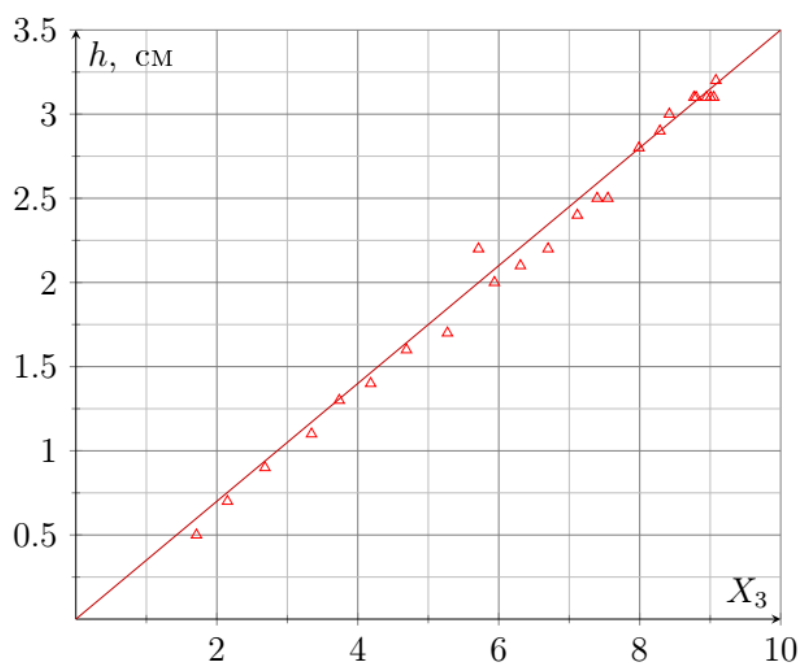
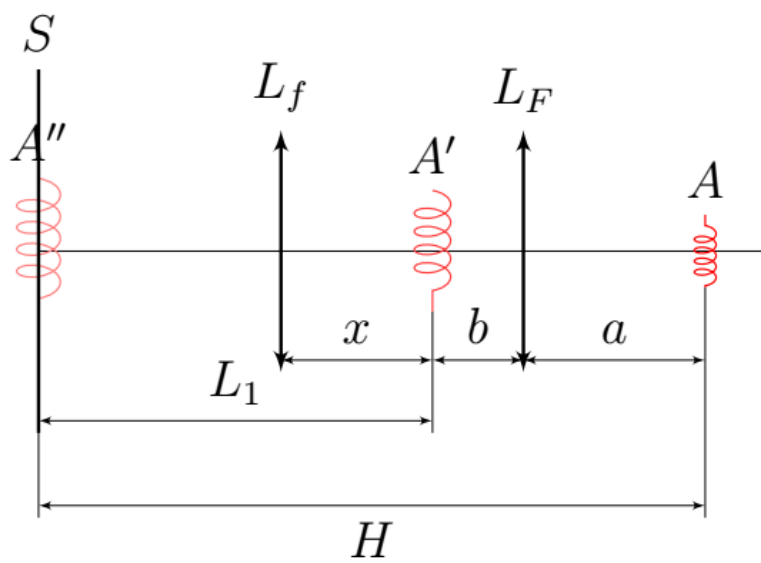
$$\frac{1}{a} + \frac{1}{b} = \frac{1}{F} \quad \Rightarrow \quad b = \frac{aF}{a - F},$$

$$\frac{1}{x} + \frac{1}{L_1 - x} = \frac{1}{f}, \quad L_1 = H - a - b$$

**We
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thin
lens
for-
mula:**

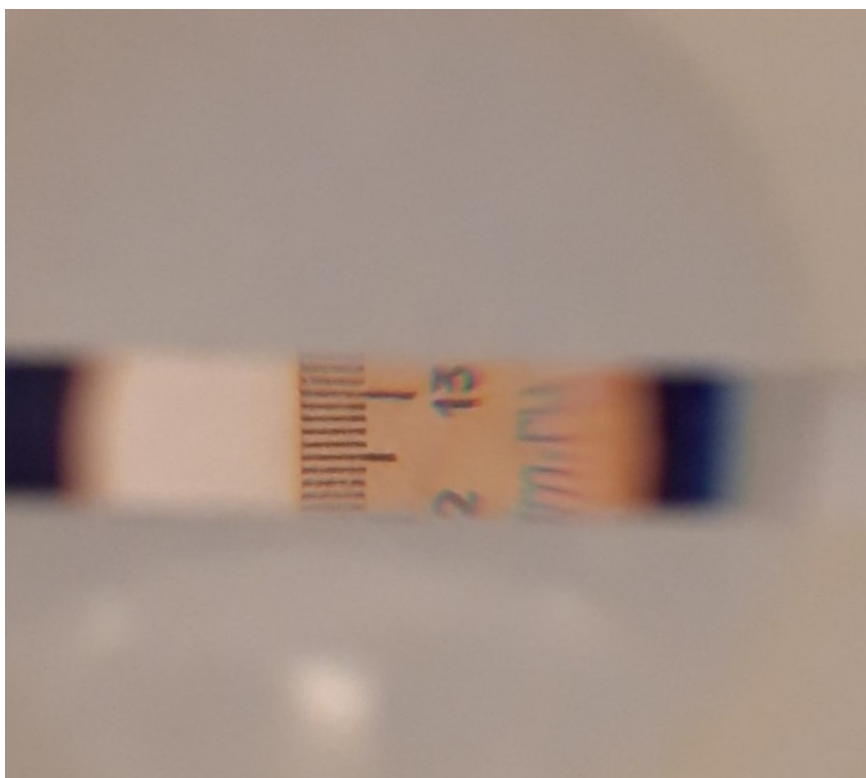
$$x^2 - xL_1 - fL_1 = 0, \quad \mathcal{D} = L_1^2 - 4fL_1, \quad x_{1,2} = L_1 \frac{1 \pm \sqrt{1 - \frac{4f}{L_1}}}{2}.$$

**With
a
smaller
 x
the
im-
age
is
larger,
so
we
choose
the
root
with
a
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nus.
Now
let's
ex-
press
the
height**

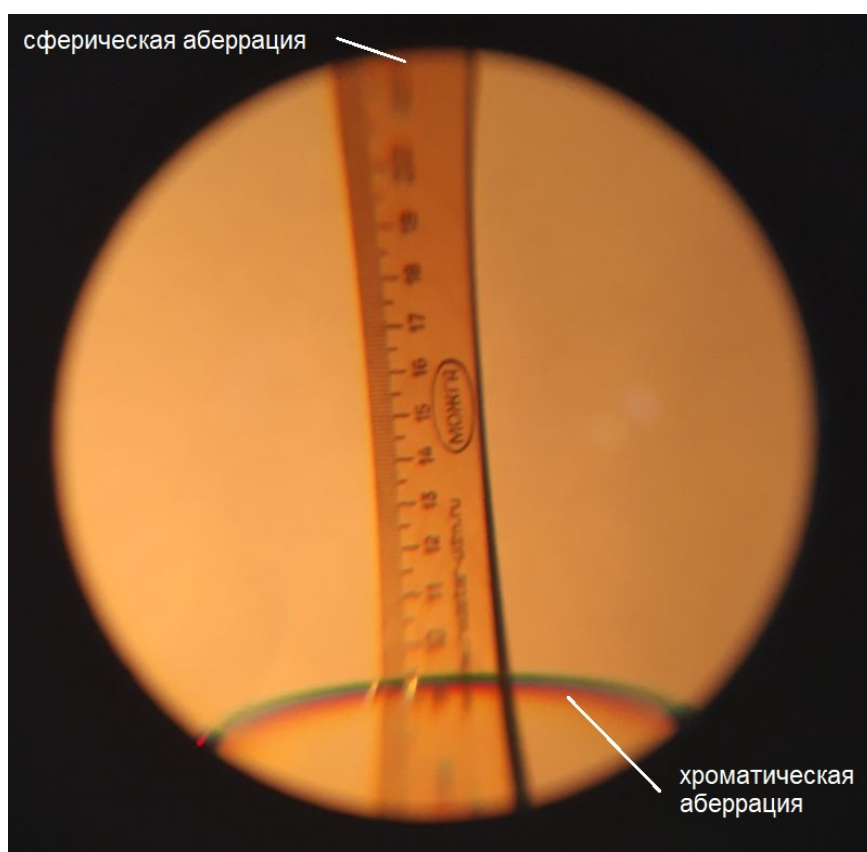


E3 Adjust the Kepler telescope so that the distance between the lens L_f and the ruler is $H \approx 1.20$ m, and through the telescope you can clearly see the magnified image of the ruler. Make the necessary measurements and calculate the experimental magnification value Γ_K of the Kepler telescope in such a system.

The magnification of the Kepler telescope can be estimated by comparing the visible image of the ruler with another known size. For example, stick two strips of masking tape onto the plastic frame of the lens. The



- E4** Observe the distortion of the ruler image at the edges of the telescope's field of view. Draw an image of the ruler as seen through the Kepler telescope. Show both chromatic and spherical aberrations in the figure. **0.40**



Source: <https://pho.rs/p/3155/s>